

Unit 116 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 89.2 ac, BAF 20 variable-radius cruise on a 300 ft grid, 44 plots, 507 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	230 sq ft/ac (SE 1.5 %)	Type	Sierran mixed conifer
Trees per acre	70	CWHR size / density	5 / D (cover 94 %)
QMD	24.6 in	SDI	283 (49 % of max 573)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	230 sq ft/ac 24.9 in	BA QMD	1 sq ft/ac 9.4 in
TPA	68	TPA	2
Volume	53.6 MBF/ac (9740 cu ft)	Volume	0.7 green tons/ac (22 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~163 sq ft/ac BA	Sampling error	3.1 % at 95 % (t = 2.02, n = 44)
Species (BA)	PP 65, WF 62, DF 42, IC 36, SP 25	Stratum standard	40 % - MEETS; 44 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	6	4.9	2.7	64	0.4
12-15.9	27	10.5	12.3	362	2.0
16-19.9	57	14.7	25.9	847	4.7
20-23.9	71	12.3	32.3	1154	6.3
24-27.9	86	10.7	39.1	1571	8.6
28-31.9	70	6.4	31.8	1407	7.7
32-35.9	58	4.3	26.4	1212	6.7
36-39.9	44	2.6	20.0	985	5.4
40-43.9	29	1.4	13.2	686	3.8
44-47.9	15	0.6	6.8	352	1.9
48-51.9	21	0.7	9.5	520	2.9
52-55.9	10	0.3	4.5	253	1.4
56-59.9	5	0.1	2.3	132	0.7
60-63.9	8	0.2	3.6	218	1.2
All	507	69.7	230.5	9763	53.7

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	144	17.6	65.5	26.1	2802	15.4
WF	136	19.6	61.8	24.0	2622	14.4
DF	93	13.1	42.3	24.4	1766	9.7
IC	79	11.8	35.9	23.6	1482	8.2
SP	55	7.6	25.0	24.6	1089	6.0
All	507	69.7	230.5	24.6	9763	53.7

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 116: plots on the 300 ft grid (orange = QA flag)

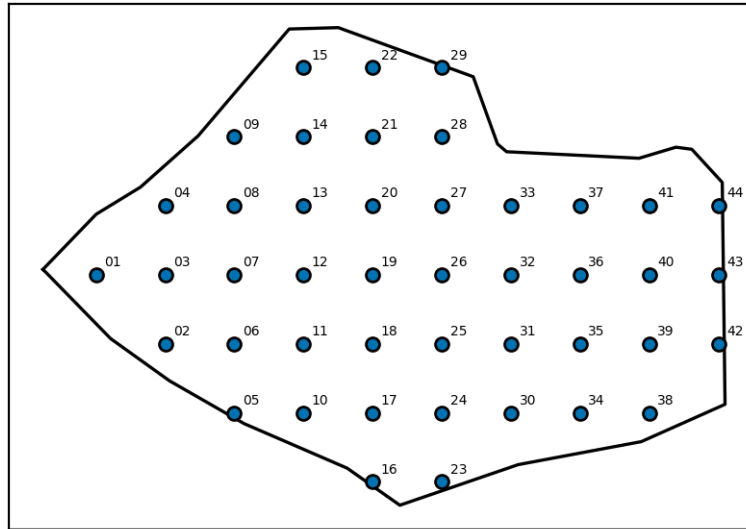


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
116-01	10	200	57	ok
116-02	12	240	57	ok
116-03	11	220	36	ok
116-04	10	200	72	ok
116-05	11	220	69	ok
116-06	12	240	67	ok
116-07	10	200	98	ok
116-08	10	200	47	ok
116-09	12	240	82	ok
116-10	11	220	63	ok
116-11	11	220	46	ok
116-12	14	280	46	ok
116-13	11	220	22	ok
116-14	11	220	21	ok
116-15	12	240	17	ok
116-16	10	200	53	ok
116-17	13	260	37	ok
116-18	11	220	52	ok
116-19	12	240	70	ok
116-20	12	240	31	ok
116-21	11	220	18	ok
116-22	10	200	17	ok
116-23	11	220	30	ok
116-24	13	260	38	ok
116-25	11	220	83	ok
116-26	11	220	68	ok
116-27	13	260	33	ok
116-28	13	260	200	ok
116-29	13	260	96	ok
116-30	11	220	94	ok
116-31	12	240	60	ok
116-32	14	280	39	ok
116-33	11	220	37	ok
116-34	10	200	18	ok
116-35	11	220	41	ok
116-36	11	220	32	ok
116-37	12	240	74	ok
116-38	13	260	28	ok
116-39	12	240	32	ok
116-40	13	260	39	ok
116-41	9	180	10	ok
116-42	11	220	44	ok
116-43	13	260	9	ok
116-44	12	240	53	ok